

AerE 421 - Lab Assignment 2

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Introduction – Part 1

For this week's lab, we were assigned two separate exercises. First, we modeled into Ansys the main spar of an aircraft subjected to a lift and drag force to collect the y and z displacements, bending stresses, point stresses, and axial stresses. It is assumed that the spar takes the entire uniform distributed load of 250 N/m. The spar is modeled as an I-beam fixed at one end and free at the other with dimensions length $L = 2.5\text{m}$, depth $d = 0.09\text{m}$, width $b = 0.05\text{m}$, web thickness $t_w = 0.001\text{m}$, and flange thickness $t_f = 0.002\text{m}$. The beam is made of an isotropic linear elastic material with Young's modulus $E = 70\text{ GPa}$ and Poisson's ratio $\nu = 0.3$.

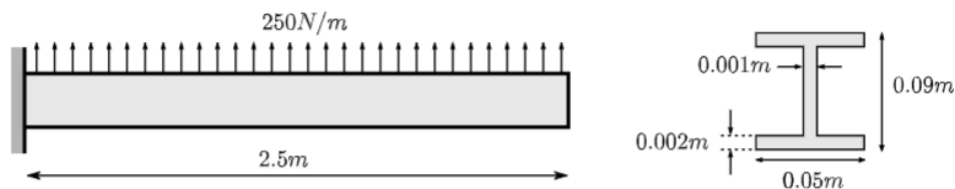


Figure 1. Dimensions and load distribution of modeled main spar of an aircraft.

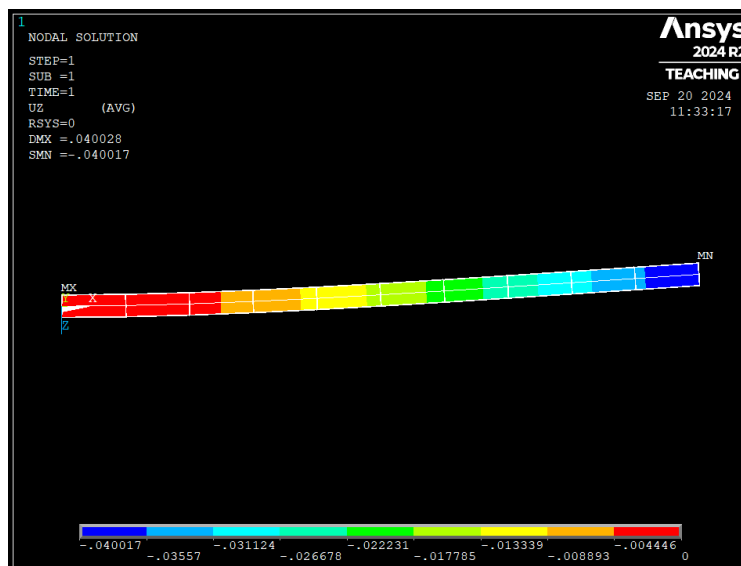


Figure 2. Top view of contour plot of the beam; Z displacement of beam (m)

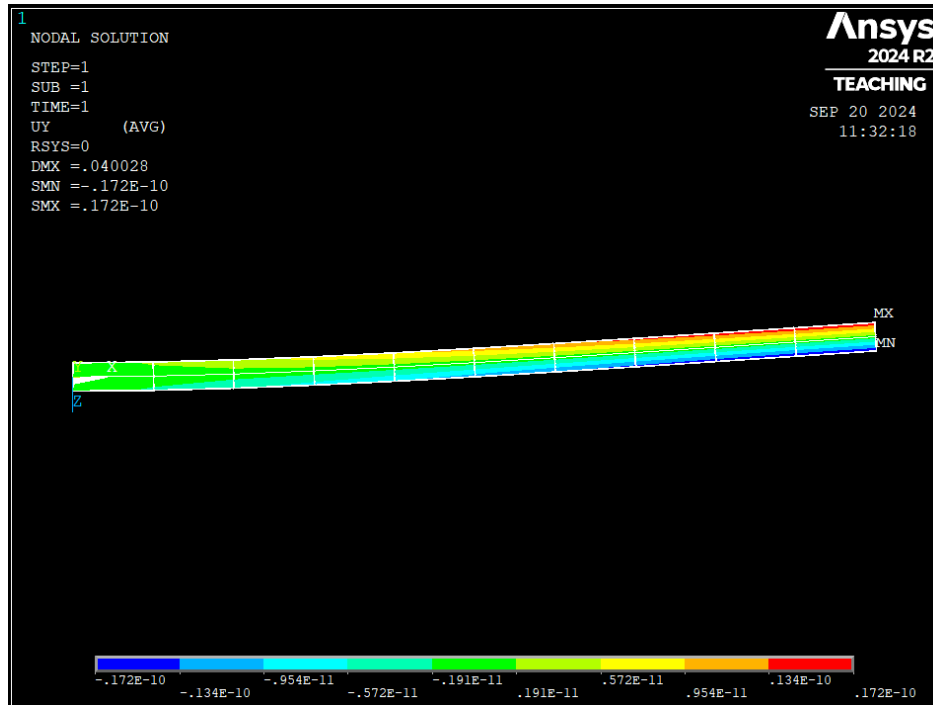


Figure 3. Front view of contour plot of the beam; Y displacement of the beam (m)

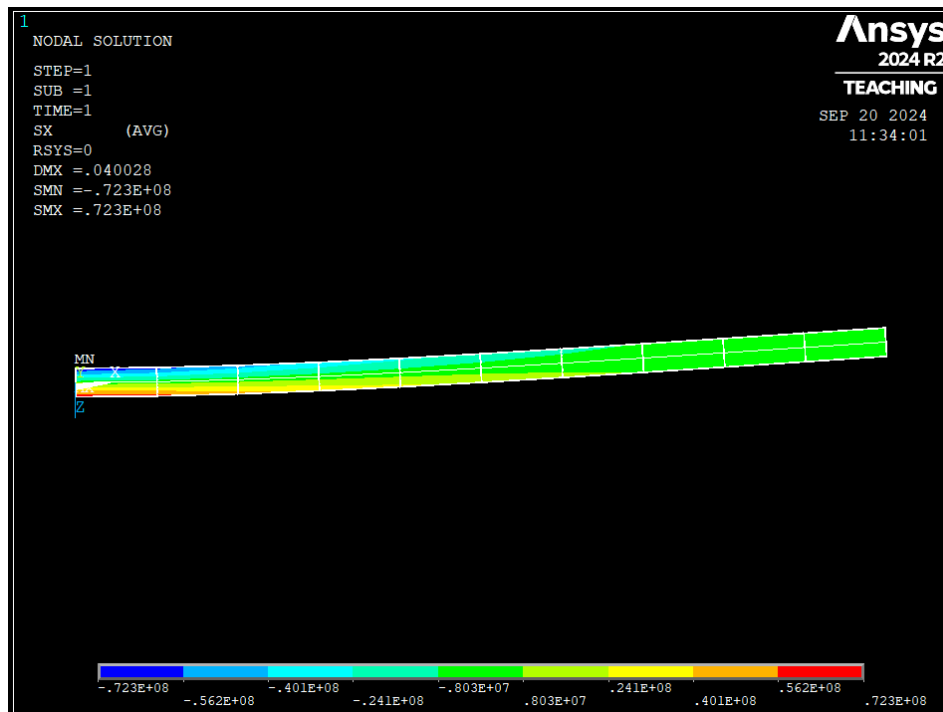


Figure 4. Top view of contour plot of the beam; X component of stress (N/m²)

Table 1. Table of element stresses, axial stresses, and bending stresses

Element n ^o	Element Stress (N/m ²)	Axial Stress (N/m ²)	Bending Stress (N/m ²)
1	-72265448	0.0000	0.19931e-8
2	-57892212	0.0000	0.15967e-8
3	-45116000	0.0000	0.12444e-8
4	-33936812	0.0000	0.93605e-9
5	-24354654	0.0000	0.67177e-9
6	-16369522	0.0000	0.45152e-9
7	-9981416	0.0000	0.27533e-9
8	-5190336	0.0000	0.14317e-9
9	-0.19963e7	0.0000	0.55071e-10
10	-0.39926e6	0.0000	0.11019e-10

Discussion

To solve for the Euler-Bernoulli beam theory, the moment of inertia was calculated $I = 4.403e-7 \text{ m}^4$. With an online beam calculator that uses the Euler-Bernoulli beam theory as the solver, we could collect displacement values to compare to those output by Ansys, resulting in the following graph:

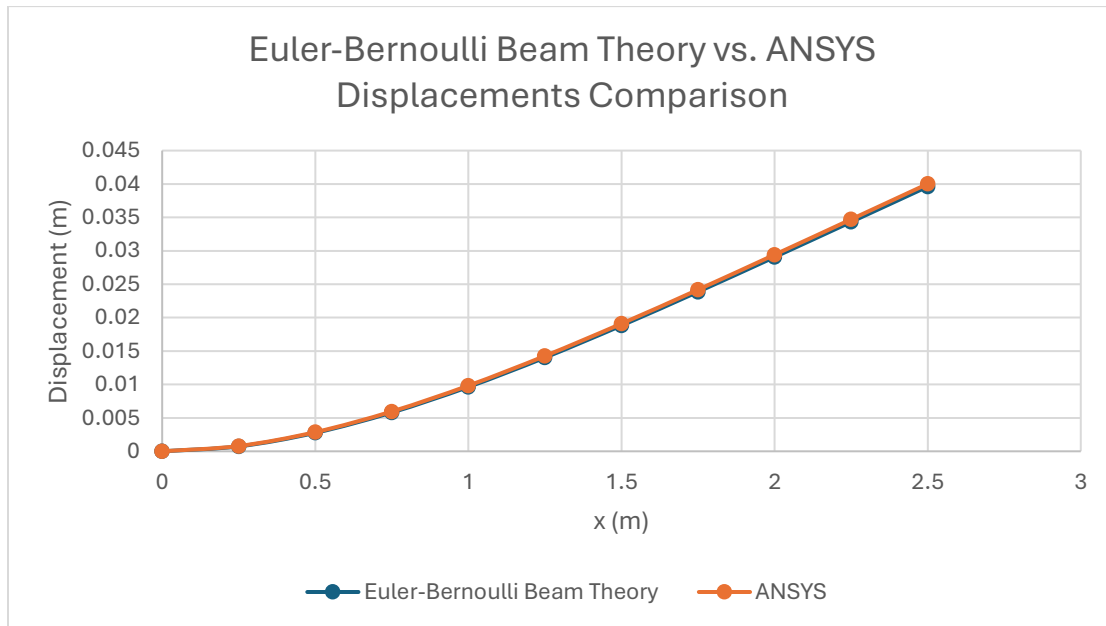


Figure 5. Euler-Bernoulli Beam Theory vs. ANSYS Displacements Comparison Graph

It becomes evident from observing the graph that the ANSYS solution and the results of the Euler-Bernoulli beam theory output the same result.

Introduction – Part 2

For part 2, we were prompted to model a landing gear with applied load at the wheels in ANSYS. We modeled a vertical main member OAB and two two-force members AC and AD. As seen in the schematic below, point B acted as a fixed end, points C and D acted as pinned joints, and point A connected all three beams as a pinned joint.

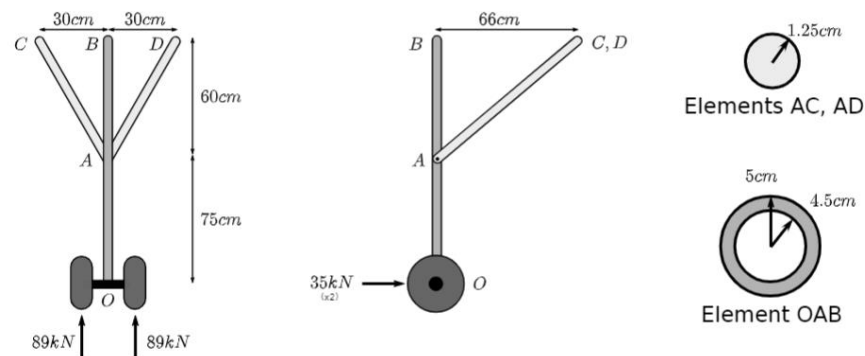


Figure 6. Dimensions and applied loads for modeled landing gear

Collected Data

PRINT U NODAL SOLUTION PER NODE					PRINT ELEMENT TABLE ITEMS PER ELEMENT		
***** POST1 NODAL DEGREE OF FREEDOM LISTING *****					***** POST1 ELEMENT TABLE LISTING *****		
LOAD STEP=	1	SUBSTEP=	1		STAT	CURRENT	CURRENT
TIME=	1.0000	LOAD CASE=	0		ELEM	MEMFORCE	AXSTRESS
THE FOLLOWING DEGREE OF FREEDOM RESULTS ARE IN THE GLOBAL COORDINATE SYSTEM					43	-178000.	-0.11938E+009
NODE	UX	UY	UZ	USUM	44	-178000.	-0.11938E+009
1	0.0000	0.44761E-003	0.53164E-001	0.53166E-001	45	-178000.	-0.11938E+009
2	0.0000	0.40436E-003	0.46626E-001	0.46627E-001	46	-178000.	-0.11938E+009
3	0.0000	0.36111E-003	0.40171E-001	0.40173E-001	47	-178000.	-0.11938E+009
4	0.0000	0.31786E-003	0.33887E-001	0.33888E-001	48	-178000.	-0.11938E+009
5	0.0000	0.27461E-003	0.27856E-001	0.27857E-001	49	-178000.	-0.11938E+009
6	0.0000	0.23135E-003	0.22164E-001	0.22165E-001	50	-178000.	-0.11938E+009
7	0.0000	0.18810E-003	0.16895E-001	0.16896E-001	51	-178000.	-0.11938E+009
8	0.0000	0.14485E-003	0.12135E-001	0.12136E-001	52	-178000.	-0.11938E+009
9	0.0000	0.10160E-003	0.79670E-002	0.79677E-002	53	-7765.3	-0.52078E+007
10	0.0000	0.58347E-004	0.44768E-002	0.44772E-002	54	-7765.3	-0.52078E+007
11	0.0000	0.15095E-004	0.17488E-002	0.17488E-002	55	-7765.3	-0.52078E+007
12	0.0000	0.13585E-004	0.33337E-003	0.33365E-003	56	-7765.3	-0.52078E+007
13	0.0000	0.12076E-004	0.61267E-003	0.61279E-003	57	-7765.3	-0.52078E+007
14	0.0000	0.10566E-004	0.11620E-002	0.11620E-002	58	-7765.3	-0.52078E+007
15	0.0000	0.90570E-005	0.13872E-002	0.13872E-002	59	-7765.3	-0.52078E+007
16	0.0000	0.75475E-005	0.13610E-002	0.13610E-002	60	-7765.3	-0.52078E+007
17	0.0000	0.60380E-005	0.11559E-002	0.11559E-002	61	-7765.3	-0.52078E+007
18	0.0000	0.45285E-005	0.84466E-003	0.84467E-003	62	-7765.3	-0.52078E+007
19	0.0000	0.30190E-005	0.49983E-003	0.49984E-003	63	-0.13350E+006	-271896889.
20	0.0000	0.15095E-005	0.19407E-003	0.19408E-003	64	-0.13350E+006	-271896889.
21	0.0000	0.0000	0.0000	0.0000	MINIMUM VALUES		
31	0.0000	0.0000	0.0000	0.0000	ELEM	44	63
41	0.0000	0.0000	0.0000	0.0000	VALUE	-0.17800E+006	-0.27190E+009
MAXIMUM ABSOLUTE VALUES					MAXIMUM VALUES		
NODE	0	1	1	1	ELEM	53	57
VALUE	0.0000	0.44761E-003	0.53164E-001	0.53166E-001	VALUE	-7765.3	-0.52078E+007

Figure 7: The first table details each element displacement, while the second displays the member forces and axial stresses present in all members of the system.

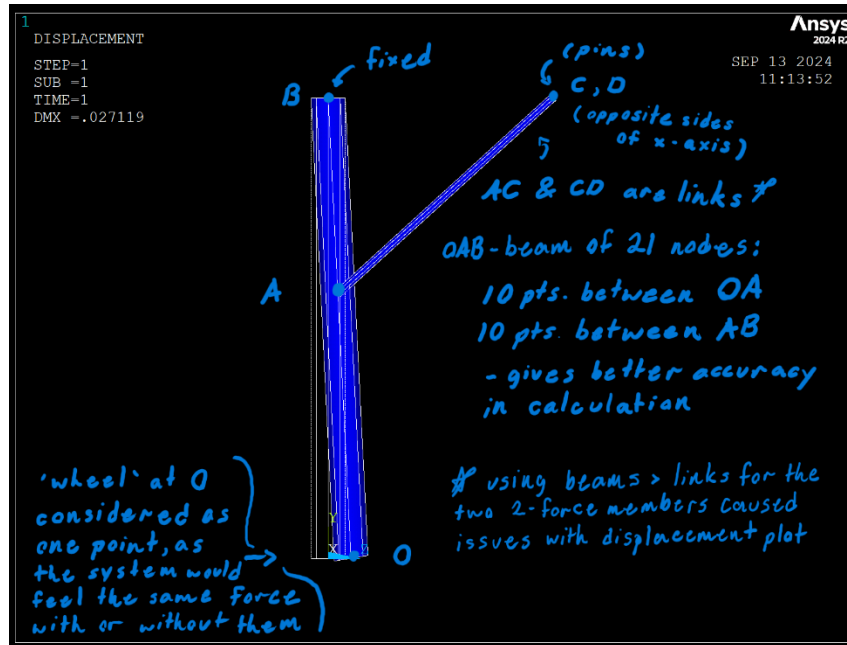


Figure 8: Side view of deformed model with design choices (expanded upon below)

Discussion

For the member OAB, we modeled it as a series of 20 two-node beam elements with a hollow-circular cross section of inner radius $r_i = 0.045$ m and outer radius $r_o = 0.05$ m. For members AC and AD, we modeled two links between the nodes A and C, and A and D. We inserted an area $A = 4.90874e-4 \text{ m}^2$, derived from the area of a circle with radius $r = 0.0125$ m.

By means of simplification, the wheels were modeled as a single point O. Thus, the forces acting on them were summed into two resultant vectors in the Y and Z directions, $F_y = 178,000$ N and $F_z = -70,000$ N. These summed forces were applied as point loads at O.

Taking the 'max' value from the table of element stresses and using the given yield strength of 1482 MPa allows us to calculate the factor of safety of the system using:

$$FOS = \frac{Y_s}{\sigma} = \frac{1482 \cdot 1000000}{|-0.27190 \cdot 10^9|} = 5.45$$

This seems a bit high for our field of study, which may be due to some issues we had during the lab seeing elements disappear after we modelled them, which is why the forces and stresses table has elements in the 60s.

Log File – Part 1

/BATCH

! /COM,ANSYS RELEASE 2024 R2 BUILD 24.2 UP20240603 10:37:18

/input,menust,tmp,"

! /GRA,POWER

! /GST,ON

! /PLO,INFO,3

! /GRO,CURL,ON

! /CPLANE,1

! /REPLOT,RESIZE

WPSTYLE,,,,,,,,0

/PREP7

!*
.

ET,1,BEAM188

!*
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!*
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MPDATA,EX,1,,70e9

MPDATA,PRXY,1,,0.3

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SECOFFSET, CENT

SECDATA,0.05,0.05,0.09,0.002,0.002,0.001,0,0,0,0,0,0

SECTYPE, 1, BEAM, I, , 0

SECOFFSET, CENT

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! /REP,FAST
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FLST,2,1,1,ORDE,1
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!*
/GO
D,P51X,ALL,,,,,,,,,
FLST,2,10,2,ORDE,2
FITEM,2,1
FITEM,2,-10
SFBEAM,P51X,1,PRES,250,250,,,,,0
FINISH
/SOL
!/STATUS,SOLU
SOLVE
FINISH
/POST1
SET,LAST
SET,LAST
! PLDISP,1
!*
!/EFACET,1
! PLNSOL, U,SUM, 0,1.0
! SAVE, lab2,db,U:\Desktop\
!/COM,ANSYS RELEASE 2024 R2      BUILD 24.2   UP20240603   16:12:30
!/REPLOT,RESIZE
! LGWRITE,'lab2_part1_log','U:\Desktop\',COMMENT
```


Log File – Part 2

/BATCH

! /COM,ANSYS RELEASE 2024 R2 BUILD 24.2 UP20240603 11:45:21

/input,start,ans,'C:\Program Files\ANSYS Inc\v242\ANSYS\apdl\'

!*
.

/PREP7

!*
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ET,1,BEAM188

!*
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ET,2,LINK180

!*
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!*
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MPDATA,PRXY,1,,0.3

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SECOFFSET, CENT

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! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

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!*
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SECDATA,4.91e-4,

SECCONTROL,0,0

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!*
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!*
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TYPE, 1

MAT, 1

REAL,

ESYS, 0

SECNUM, 1

TSHAP,LINE

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FITEM,2,2

E,P51X

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MAT, 1
REAL,
ESYS, 0
SECNUM, 1
TSHAP,LINE
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EGEN,20,1,P51X,,,,,,,,,
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ESYS, 0
SECNUM, 2

TSHAP,LINE

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FITEM,2,31

E,P51X

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FITEM,2,11

FITEM,2,41

E,P51X

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! /ESHAPE,1

! /EFACET,1

! /RATIO,1,1,1

/CFORMAT,32,0

! /REPLOT

!*
!

! EPLOT

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! /REP,FAST
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! /DIST, 1 ,0.924021,1

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! /REP,FAST
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! /DIST, 1 ,0.924021,1
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! /REP,FAST
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! /DIST, 1 ,0.924021,1
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! /REP,FAST
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! /DIST, 1 ,0.924021,1
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! /REP,FAST

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! /DIST, 1 ,0.924021,1
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! /REP,FAST
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! /DIST, 1 ,0.924021,1
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! /REP,FAST
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! /DIST, 1 ,0.924021,1
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! /REP,FAST
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! /DIST, 1 ,0.924021,1
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! /REP,FAST
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! /DIST, 1 ,0.924021,1
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! /REP,FAST
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! /DIST, 1 ,0.924021,1
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! /REP,FAST
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! /DIST, 1 ,0.924021,1

! /REP,FAST

! /DIST, 1 ,0.924021,1

! /REP,FAST

! /DIST, 1 ,0.924021,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,0.924021,1

! /REP,FAST

! /DIST, 1 ,0.924021,1

! /REP,FAST

! /DIST, 1 ,0.924021,1

! /REP,FAST

! /DIST, 1 ,0.924021,1

! /REP,FAST

! /DIST, 1 ,0.924021,1

! /REP,FAST

! /DIST, 1 ,0.924021,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,0.924021,1

! /REP,FAST

! /DIST, 1 ,0.924021,1

! /REP,FAST

! /VIEW,1,1

! /ANG,1

! /REP,FAST

! /VIEW,1,,,1

```
! /ANG,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
TYPE, 1
MAT, 1
REAL,
ESYS, 0
SECNUM, 1
TSHAP,LINE
!*
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
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! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
FLST,2,2,1
FITEM,2,1
FITEM,2,2
E,P51X
FLST,4,1,2,ORDE,1
FITEM,4,33
EGEN,10,1,P51X, , , , , , , , , ,
! /DIST, 1 ,0.924021,1
! /REP,FAST

! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
FLST,2,1,1,ORDE,1
FITEM,2,21
!*
/GO
D,P51X,ALL, , , , , , , , ,
FLST,2,2,1,ORDE,2
FITEM,2,31
FITEM,2,41

!*
.

/GO

D,P51X,UX, , , ,UY,UZ, , , ,

FLST,2,1,1,ORDE,1

FITEM,2,1

!*
.

/GO

F,P51X,FY,178000

FLST,2,1,1,ORDE,1

FITEM,2,1

!*
.

/GO

F,P51X,FZ,-70000

! /VIEW,1,,,1

! /ANG,1

! /REP,FAST

! /VIEW,1,1

! /ANG,1

! /REP,FAST

FINISH

/SOL

! /STATUS,SOLU

SOLVE

! /DIST, 1 ,0.924021,1

! /REP,FAST

! /DIST, 1 ,0.924021,1

! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
FINISH
/PREP7
FLST,2,32,2,ORDE,2
FITEM,2,11
FITEM,2,-42
EDELE,P51X
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1

! /REP,FAST
FLST,2,2,1
FITEM,2,1
FITEM,2,2
E,P51X
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1

! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
FLST,4,1,2,ORDE,1
FITEM,4,43
EGEN,10,1,P51X,,,,,,,,,
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST

! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
FLST,2,2,1
FITEM,2,11
FITEM,2,12
E,P51X
FLST,4,1,2,ORDE,1
FITEM,4,53
EGEN,10,1,P51X, , , , , , , , , ,
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1

```
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
TYPE, 2
MAT, 1
REAL,
ESYS, 0
SECNUM, 2
TSHAP,LINE
!*
! /VIEW,1,1
! /ANG,1
! /REP,FAST
! /VIEW,1,,1
! /ANG,1
! /REP,FAST
! /VIEW,1,1
! /ANG,1
! /REP,FAST
! /VIEW,1,,,1
! /ANG,1
! /REP,FAST
FLST,2,2,1
FITEM,2,41
```

```
FITEM,2,11
E,P51X
FLST,2,2,1
FITEM,2,11
FITEM,2,31
E,P51X
! EPLOT
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,0.924021,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
! /DIST, 1 ,1.082226,1
! /REP,FAST
FINISH
/SOL
! /STATUS,SOLU
SOLVE
! /VIEW,1,1
! /ANG,1
! /REP,FAST
FINISH
/POST1
! PLDISP,1
! /DIST, 1 ,1.082226,1
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! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,0.924021,1

! /REP,FAST

! /DIST, 1 ,0.924021,1

! /REP,FAST

! /DIST, 1 ,0.924021,1

! /REP,FAST

! /DIST, 1 ,0.924021,1

! /REP,FAST

! /DIST, 1 ,1.082226,1

! /REP,FAST

! /DIST, 1 ,0.924021,1

! /REP,FAST

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